

FINITE-DIMENSIONAL POSITIVE CASES FOR NET EXTENSION

RUN FA_BANACH_001

Status. `literature_implied_answer` (partial subcase). This is not a new proof and not a full solution of Naor’s classification question. It records that standard finite-Nagata Lipschitz extension theory implies a positive finite-dimensional-domain subcase.

SOURCE QUESTION

Naor’s *Uniform nonextendability from nets*, arXiv:1506.05391, proves that there are Banach spaces X, Y , a 1-net $\mathcal{N} \subset X$, and a Lipschitz map $f : \mathcal{N} \rightarrow Y$ such that no uniformly continuous extension $F : X \rightarrow Y$ exists.

Immediately after the theorem, Naor asks for an understanding of those pairs of Banach spaces X, Y for which Rosendal’s question has a positive answer: for every 1-net $\mathcal{N}$ of X , every Y -valued Lipschitz map on $\mathcal{N}$ should admit a uniformly continuous extension to X .

SUPPORTING LITERATURE

Freeman–Gartland, *Lipschitz functions on unions and quotients of metric spaces*, arXiv:2212.08707, Lemma 3.1, records that doubling spaces have finite Nagata dimension and that finite Nagata dimension is preserved under bi-Lipschitz embeddings. Lemma 3.2 states that if Z is a metric space and $S \subset Z$ is closed with finite Nagata dimension, then there is a weak-star-weak-star continuous bounded linear extension operator

$$E : \text{Lip}_0(S) \longrightarrow \text{Lip}_0(Z).$$

Aliaga–Medina, *Lipschitz extension and Lipschitz-free spaces over nets in normed spaces*, arXiv:2601.03131, Fact 2.2(c), identifies the Banach-valued extension constant $e(S, M)$ with the infimum of the constants for weak-star Lipschitz extendability. Equivalently, such an operator implies that every Banach-valued 1-Lipschitz map on S extends to an L -Lipschitz map on M , for some finite L .

Literature-implied subcase. *Let X be a finite-dimensional Banach space and let Y be any Banach space. For every 1-net $\mathcal{N} \subset X$, every Lipschitz map $f : \mathcal{N} \rightarrow Y$ has a Lipschitz, hence uniformly continuous, extension $F : X \rightarrow Y$.*

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Proof. A 1-net in Naor’s sense is uniformly discrete, hence closed in X . After identifying the finite-dimensional normed space X with $\mathbb{R}^n$, the metric space X , and therefore every subset $\mathcal{N} \subset X$, is doubling and has finite Nagata dimension.

Apply Freeman–Gartland Lemma 3.2 with $Z = X$ and $S = \mathcal{N}$. This gives a weak-star continuous bounded linear extension operator $\text{Lip}_0(\mathcal{N}) \rightarrow \text{Lip}_0(X)$. By Aliaga–Medina Fact 2.2(c), $e(\mathcal{N}, X) < \infty$. Thus every Banach-valued Lipschitz map $f : \mathcal{N} \rightarrow Y$ extends to a Lipschitz map on X . Lipschitz maps are uniformly continuous, so this is a positive case of Rosendal’s net-extension property. $\square$

SCOPE

This proves only the finite-dimensional-domain subcase of the positive-pairs classification problem. It does not address infinite-dimensional domains, and it does not settle either direction of the coarse-versus-uniform embeddability questions discussed in Naor’s introduction.

SEARCH AND PROVENANCE NOTES

The run indexes were searched for arXiv:1506.05391, the title *Uniform nonextendability from nets*, Rosendal, 1-nets, Lipschitz extension, finite Nagata dimension, weak-star Lipschitz extendability, and $e(S, M)$. No prior packet for this exact finite-dimensional-domain subcase was found. Aliaga–Medina cite Naor’s paper, but the connection to Rosendal’s positive-pairs question is an agent-identified implication rather than an explicit “we answer this question” statement.